SANGHAMITRA JOHRI

Seattle, WA | +1 (206) 387-5316 | Email | LinkedIn | GitHub

SUMMARY

ML Systems Engineer with 2+ years of experience in GPU-accelerated inference for vision and transformer models. Experienced in CUDA acceleration, kernel profiling, and benchmarking inference pipelines, with emphasis on memory-bound workloads and edge deployment efficiency.

EDUCATION

University of Washington, Seattle — Sep. 2022 – Jun. 2024
Master of Science - Electrical and Computer Engineering

Rajasthan Technical University, India — Aug. 2016 – Jun. 2020
Bachelor of Technology - Electronics and Communication Engineering

EXPERIENCE

Armada Inc. — **Feb. 2025 – present**
Artificial Intelligence Engineer, **Tech Stack: Python, Pytorch, Triton Inference Server** — Bellevue, USA

- Designed and deployed an **end-to-end real-time vision inference pipeline** with GPU-accelerated execution and post-processing
- **Re-architected object tracking** pipeline from hybrid CPU–GPU to fully GPU-resident execution by converting core modules to **PyTorch**, reducing host CPU demand by **25×** and eliminating synchronization bottlenecks
- **Benchmarked pipeline performance across single-GPU and multi-GPU configurations**, evaluating scaling efficiency, GPU saturation behavior, and cross-device workload distribution
- **Scaled multi-stream inference** from **10 to 100+ parallel video streams** on NVIDIA A100 GPUs, systematically benchmarking throughput, latency stability, and resource utilization under increasing load
- **Analyzed and optimized large-scale ML inference workloads on NVIDIA GPUs**, improving throughput by 2× and stabilizing latency while scaling from 10 to 100+ parallel streams

University of Washington — **Jul. 2024 – Jan. 2025**
Artificial Intelligence Engineer - Systems, **Tech Stack: C++, CUDA, Pytorch** — Seattle, USA

- **LLM Inference Optimization using KV-Caching**
  - Optimized inference in nanoGPT by implementing KV-caching, **achieving a 2.0x improvement** in per token generation speed
  - Benchmarked transformer implementations using NVIDIA Nsight Systems and PyTorch Profiler, analyzing key performance metrics to reduce latency and improve computational efficiency
- **Performance Profiling and Optimization of Llama Decode Layer**
  - Analyzed key components such as Flash-Attention, MLPs, and GEMM operations, generating latency and throughput plots across **batch sizes (1–64) with 3x speedup for batch inference** and sequence lengths (256–16,384) to identify scalability trends
  - Utilized NVIDIA Nsight Systems, PyTorch Profiler, and Matplotlib for GPU performance analysis, identifying bottlenecks in memory bandwidth and optimizing arithmetic intensity

Hood Tech Vision Inc. — **Jul. 2023 – Sep. 2023**
Software Engineering Intern – Systems, **Tech Stack: C++, Linux, embedded systems** — Oregon, USA

- **Designed and implemented a high-performance image processing pipeline** that dynamically loads and applies user-defined filters, improving object detection accuracy by 35% for autonomous perception systems
- Built a multi-threaded C++ pipeline using Linux system calls, achieving a 3× throughput improvement on high-resolution video frames
- Implemented runtime dynamic loading (dlopen/dlsym) of user-defined C++ filters, enabling flexible deployment and compiler–runtime adaptability
- Collaborated with embedded and systems engineers to ensure real-time performance constraints were met

Newspace Research and Technology — **Nov. 2021 – Aug. 2022**
Artificial Intelligence Engineer, **Tech Stack: Python, Model Optimization, Pytorch** — Bangalore, India

- **Led end-to-end delivery of UAV perception systems**, coordinating labeling automation, model training, and deployment
- Implemented SiamMask and ByteTrack for small-object tracking in EO/IR imagery, supporting autonomous aerial operations
- Fine-tuned YOLO v4 models to achieve a **40% increase in mAP@0.75** by applying hyperparameter optimization techniques and monitoring GPU performance on various NVIDIA low-power devices for efficient edge deployment
- Optimized models using pruning and 8-bit quantization to ensure efficient GPU performance on low-power devices like NVIDIA Jetson Tx2

PUBLICATION

Sensory Data Classification for Gait Assessment using Deep Neural Networks: A Comparative Study with SGD and Adam Optimizer, **IEEE Interdisciplinary Approaches in Technology and Management for Social Innovation - 2024 (link)**

LSTM neural network-based classification of sensory signals for healthy and unhealthy gait assessment, **Predictive Modeling in Biomedical Data Mining and Analysis, Elsevier 2022 (link)**

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Embedded C

ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV

Cloud & DevOps Tools: Apache Flink, Kubernetes, Docker, GCP, AWS, Git, Jira

GPU Programming & Perf. Tools: CUDA, NVIDIA Nsight, PyTorch Profiler